DESIGN CODE OF CLA

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/cla8.vhd [VHDL]

```
1  ----------------------------------------------------------------------------------
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date: 08/13/2026 04:46:52 PM
6  -- Design Name:
7  -- Module Name: cla8 - structural
8  -- Project Name:
9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 ----------------------------------------------------------------------------------
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity cla8 is
35     Port ( a : in STD_LOGIC_VECTOR (7 downto 0);
36            b : in STD_LOGIC_VECTOR (7 downto 0);
37            cin : in std_logic;
38            sum : out STD_LOGIC_VECTOR (7 downto 0);
39            cout : out STD_LOGIC);
40 end cla8;
41
42 architecture structural of cla8 is
43
44     signal c : std_logic_vector(7 downto 0);
45
46     signal p : std_logic_vector(7 downto 0);
```

```
    signal g : std_logic_vector(7 downto 0);

begin

    p <= a xor b;
    g <= a and b;


    c(0) <= cin;
    c(1) <= g(0) or (p(0) and (cin));
    c(2) <= g(1) or (p(1) and (g(0) or (p(0) and (cin))));
    c(3) <= g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin)))));
    c(4) <= g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin)))))));
    c(5) <= g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))))));
    c(6) <= g(5) or (p(5) and (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin)))))))));
    c(7) <= g(6) or (p(6) and (g(5) or (p(5) and (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))))))));
    cout <= g(7) or (p(7) and (g(6) or (p(6) and (g(5) or (p(5) and (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin)))))))));

    sum(0) <= a(0) xor b(0) xor c(0);
    sum(1) <= a(1) xor b(1) xor (g(0) or (p(0) and (cin)));
    sum(2) <= a(2) xor b(2) xor (g(1) or (p(1) and (g(0) or (p(0) and (cin))));
    sum(3) <= a(3) xor b(3) xor (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin)))));
    sum(4) <= a(4) xor b(4) xor (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))));
    sum(5) <= a(5) xor b(5) xor (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin)))));
    sum(6) <= a(6) xor b(6) xor (g(5) or (p(5) and (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))));
    sum(7) <= a(7) xor b(7) xor (g(6) or (p(6) and (g(5) or (p(5) and (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin)))))));

end structural;
```

TEST BENCH CODE OF CLA

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/tb/cla8_tb.vhd [VHDL]

```
----------------------------------------------------------------------------------
-- Company:
-- Engineer:
--
-- Create Date: 08/13/2026 05:08:58 PM
-- Design Name:
-- Module Name: cla8_tb - testing
```

```
-- Project Name:
-- Target Devices:
-- Tool Versions:
-- Description:
--
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity cla8_tb is
--  Port ( );
end cla8_tb;

architecture testing of cla8_tb is

    component cla8 is
        Port ( a : in STD_LOGIC_VECTOR (7 downto 0);
                b : in STD_LOGIC_VECTOR (7 downto 0);
                cin : in std_logic;
                sum : out STD_LOGIC_VECTOR (7 downto 0);
                cout : out STD_LOGIC);
    end component;


    signal a : std_logic_vector(7 downto 0);
    signal b : std_logic_vector(7 downto 0);
    signal s : std_logic_vector(7 downto 0);
    signal cin : std_logic;
    signal cout : std_logic;



begin

```

```
59
60      cla_test : cla8 port map(
61          a => a,
62          b => b,
63          cin => cin,
64          sum => s,
65          cout =>cout
66      );
67
68
69      process
70
71      begin
72          cin <= '0';
73          a <= "01010101";
74          b <= "10101010";
75
76          wait for 10ns;
77
78          a <= "00110011";
79          b <= "11001100";
80
81          wait for 10ns;
82
83          a <= "10101010";
84          b <= "11001100";
85
86          wait for 10ns;
87
88          cin <= '1';
89          a <= "00110001";
90          b <= "10011100";
91
92          wait for 10ns;
93
94          a <= "11001010";
95          b <= "11000011";
96
97          wait for 10ns;
98
99          a <= "11111111";
100         b <= "11111111";
101
102         wait for 10ns;
103
104         cin <= '0';
105         a <= "00000000";
106         b <= "00000000";
107
108         wait for 10ns;
109
```

```
110         a <= "01011010";
111         b <= "11110011";
112 
113         wait for 10ns;
114 
115         cin <= '1';
116         a <= "00000001";
117         b <= "10000111";
118 
119         wait for 10ns;
120 
121         a <= "10101010";
122         b <= "11111111";
123 
124 
125         wait;
126 
127     end process;
128 
129 
130 
131 end testing;
```

SCREENSHOT OF CODE

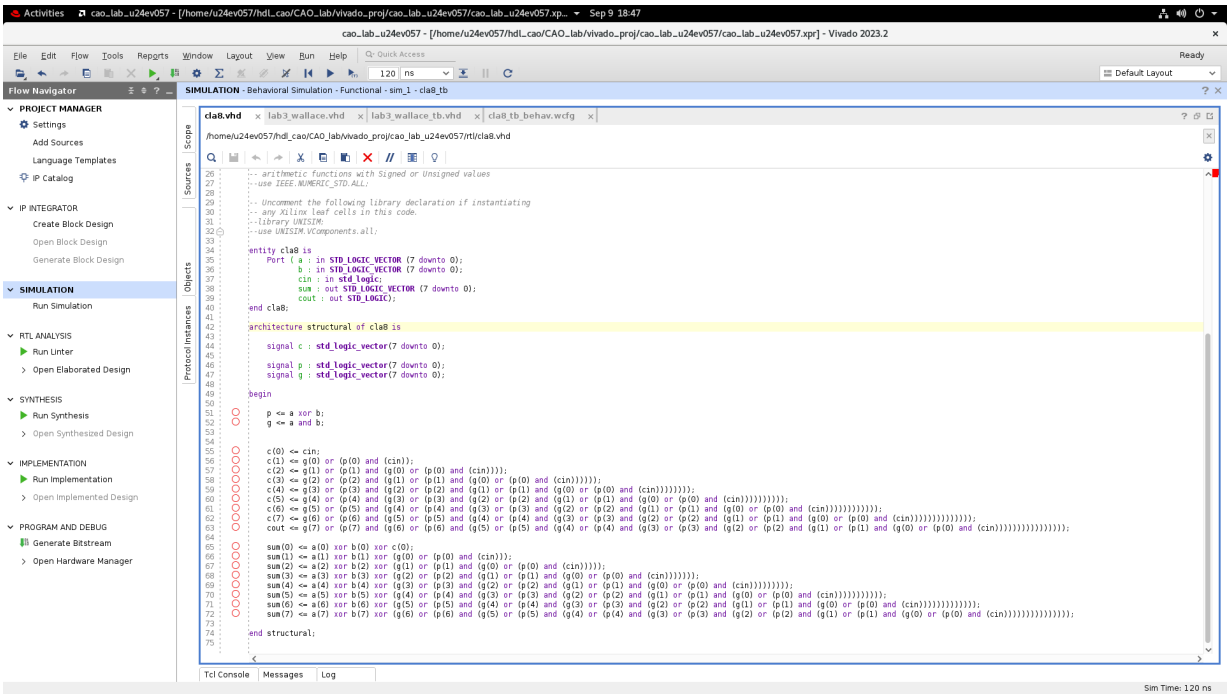


Figure 1: Design

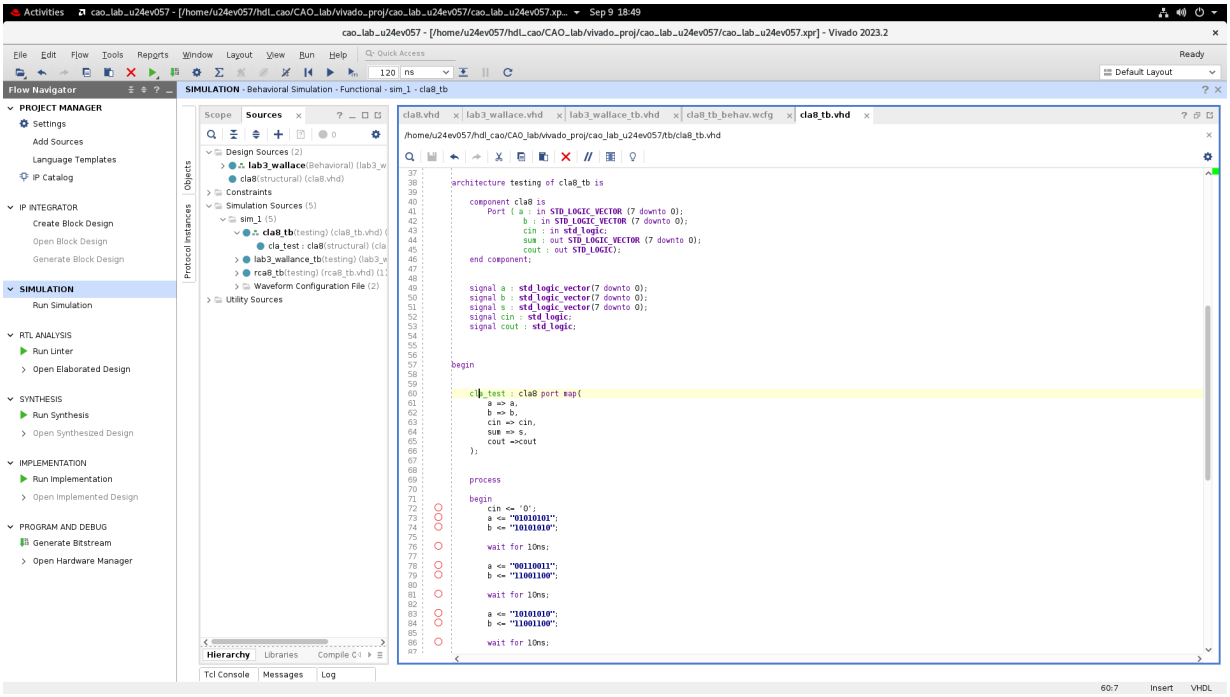


Figure 2: Tb

WAVEFORM

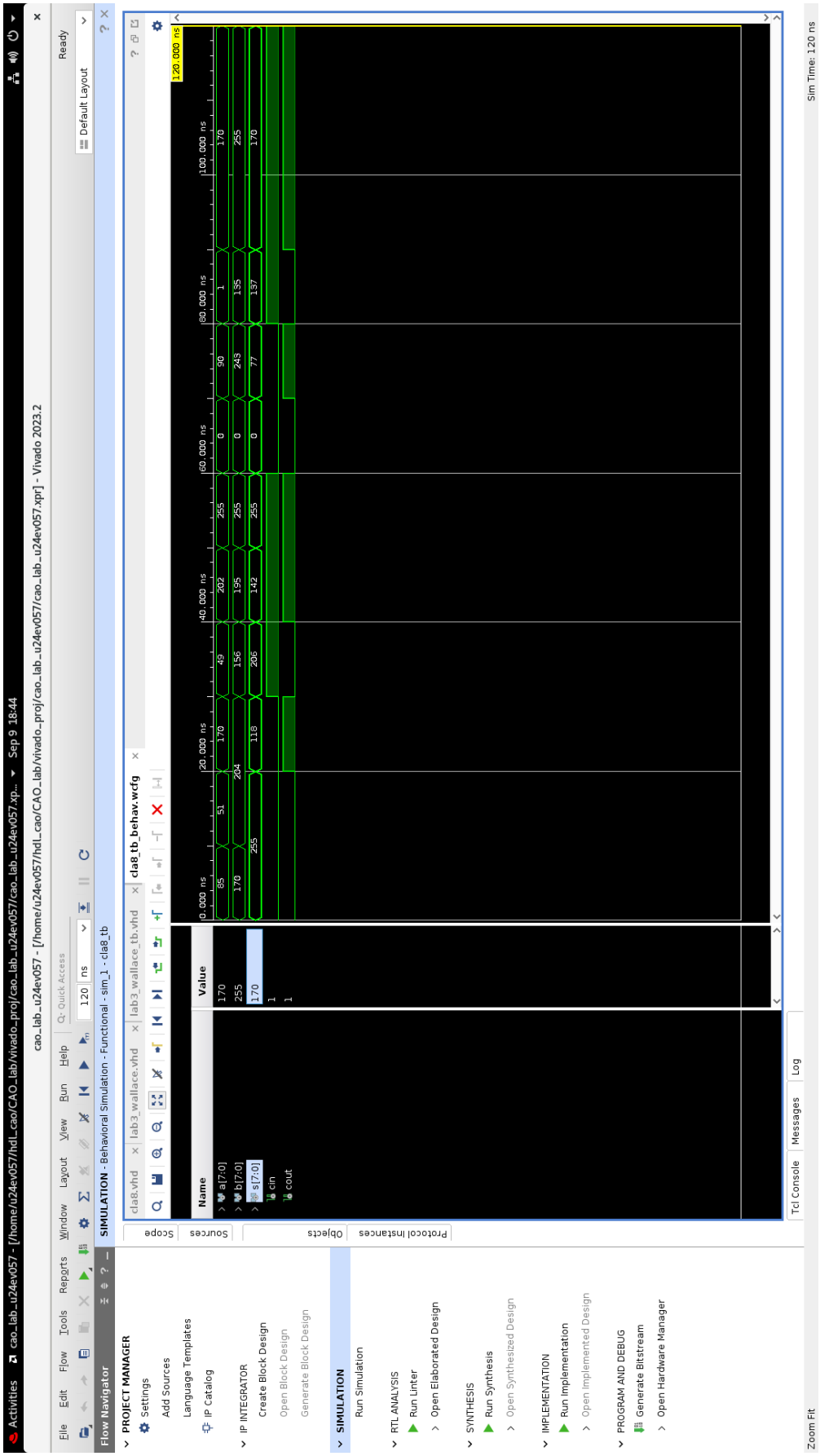


Figure 3: Waveform